

Quiz 2: September 22

Name: Solutions

SID: _____

Problem 1. Does there exist a linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$T(1, 1, 0) = 4$$

$$T(0, 1, 1) = 3$$

$$T(3, 0, -3) = 1?$$

Give an example or prove such a T cannot exist.

$$\begin{pmatrix} 3 \\ 0 \\ -3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow T \begin{pmatrix} 3 \\ 0 \\ -3 \end{pmatrix} = 3T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 3T \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{but LHS} = 1$$

$$\text{RHS} = 3. \quad \sum \nabla$$

Problem 2. Suppose that $T : \mathbb{C}^5 \rightarrow \mathbb{C}^3$ is a linear map such that there exists a vector $v \in \mathbb{R}^3$ with $Tv \neq 0$. What are the possible values for $\dim \text{null } T$? Give examples to show each claimed value is possible.

$$5 = \dim(\text{Im}(T)) + \dim(\text{Null}(T))$$

$$1 \leq \dim(\text{Im}(T)) \leq 3$$

$$\dim(\text{Null}(T)) \geq 2$$

So pairs

$$(3, 2)$$

$$(2, 3)$$

$$(1, 4)$$

$$\rightarrow T(x, y, z, a, b) = (x, y, z)$$

$$= (x, y, 0)$$

$$= (x, 0, 0).$$

Concept Map.

